

A dimension-independent collision-depth invariant for right-corner orthoscheme reflection groups

Closed-form growth rate $r_n = 1 + 2 \cos(2\pi/(n+3))$ and unconditional Class C faithfulness in every dimension

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Abstract

For every $n \geq 3$ the reflection group W_n generated by the $n+1$ face-reflections of an n -dimensional right-corner orthoscheme admits a positional combinatorial invariant $\text{cd}_n(a_1, \dots, a_n) \in \{\infty, 4, 3\}$, the *collision-depth*, determined entirely by two integer statistics (e, t) of the leg sequence: the number of endpoint-adjacent equal pairs and the number of consecutive equal triples (Theorem 9). The same group has Coxeter–Steinberg growth rate $r_n = 1 + 2 \cos(2\pi/(n+3))$ in closed form (Theorem 2), and its affine representation $\rho_n : W_n \rightarrow \text{Aff}(\mathbb{R}^n)$ is unconditionally faithful on every leg tuple with pairwise distinct positive rational coordinates (Theorem 8). The faithfulness proof rests on a uniform Schur-complement determinant identity $\det Q_n = -\prod_i a_i^2$ (Lemma 6) together with rank-0 Mordell descent on three Diophantine quartics whose Jacobians are Cremona curves 24a1 and 72a2.

1. Setting

Fix $n \geq 2$. An n -dimensional right-corner orthoscheme $O = O(a_1, \dots, a_n)$ is the simplex in $\mathbb{R}^n$ with vertex sequence

$$V_0 = 0, \quad V_k = V_{k-1} + a_k e_k \quad (1 \leq k \leq n),$$

where $e_1, \dots, e_n$ is the standard basis and $(a_1, \dots, a_n) \in \mathbb{Q}_{>0}^n$. The $n+1$ bounding hyperplanes carry reflections $R_0, R_1, \dots, R_n$; we write $W_n = W(O) := \langle R_0, \dots, R_n \rangle \leq \text{Isom}(\mathbb{R}^n)$. The orbit of O under W_n is explored by breadth-first search in the Cayley graph relative to $\{R_0, \dots, R_n\}$, and

$$\mathcal{O}_d(O) = \{g \cdot O : g \in W_n, \text{ minimal word-length } d\}, \quad u_d^n(O) := |\mathcal{O}_d(O)|, \quad u_0 = 1.$$

All distances are computed in exact rational arithmetic.¹

Independence of legs. A symbolic computation in $\mathbb{Q}(a_1, \dots, a_n)$ shows that the perpendicular face-pair structure of O is independent of the leg values: pairs (i, j) with $|i - j| \geq 2$ commute identically, while pairs $(i, i+1)$ have a leg-dependent dihedral angle with $m_{i,i+1} = \infty$ for generic legs. The abstract Coxeter group is therefore the right-angled Coxeter group on the path diagram P_{n+1} :

$$W_n \cong \langle R_0, \dots, R_n \mid R_i^2 = 1, (R_i R_j)^2 = 1 \text{ for } |i - j| \geq 2 \rangle, \quad (1)$$

the same group for every nondegenerate orthoscheme of dimension n . This is the standard right-angled Coxeter system on the commutation graph that is the complement of P_{n+1} .

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¹A companion paper treats the dimension- $n = 2$ case in detail, including a universality theorem and eight length-10 affine relations specific to planar right triangles, none of which generalise to $n \geq 3$. The present paper is self-contained and does not depend on that companion.

Steinberg upper bound. Steinberg’s word-growth formula (see Humphreys [1], §5.12) gives $1/W_n(t^{-1}) = \sum_I (-1)^{|I|}/W_I(t)$, summed over cliques I of the commutation graph G_n (cliques correspond to subsets of pairwise-commuting generators, hence to finite parabolics $W_I \cong (\mathbb{Z}/2)^{|I|}$). The number of k -cliques of G_n is $\binom{n-k+2}{k}$, so substituting $W_I(t) = (1+t)^{|I|}$:

$$\frac{1}{W_n(t^{-1})} = \sum_{k \geq 0} (-1)^k \binom{n-k+2}{k} \frac{1}{(1+t)^k}. \quad (2)$$

For every faithful or non-faithful affine representation of W_n one has $u_d^n \leq [t^d] W_n(t)$. The aim of §§2–4 is (a) to identify $W_n(t)$ in closed form, (b) to certify equality $u_d^n = [t^d] W_n(t)$ unconditionally for distinct positive rational legs, and (c) to read off the depth at which the inequality first becomes an equality (the collision-depth).

Remark 1 (Scope, and relation to the planar case). This is a companion note. The planar case $n = 2$ (where the group is $V_4 * \mathbb{Z}/2$, the metric is the lamplighter-type formula, and the growth series is transcendental) is treated in the main paper and is not reproved here; what follows is the part of the picture that is uniform in the dimension. Three things are established: the closed form for r_n (§2), the uniform Schur-complement determinant identity (§1), which is what makes Class C faithfulness unconditional in every dimension (§1), and the dimension-independent collision-depth invariant (§4). Two things are deliberately *not* attempted: the transcendence questions, which are special to $n = 2$; and any statement about non-right-corner orthoschemes.

2. The closed-form growth rate r_n

Theorem 2 (Closed-form n -dim Coxeter rate). *For every $n \geq 2$, the abstract orbit-growth rate of W_n is*

$$r_n = 1 + 2 \cos\left(\frac{2\pi}{n+3}\right).$$

Proof. Start from Steinberg’s identity (2). Substitute $y = -1/(1+t)$, so that $(1+t)^{-k} = (-y)^k = (-1)^k y^k$ and the two $(-1)^k$ factors cancel:

$$\frac{1}{W_n(t^{-1})} = \sum_{k \geq 0} \binom{n-k+2}{k} y^k =: f_n(y).$$

The polynomial $f_n(y)$ is the standard Fibonacci–Pell polynomial: writing $P_0(y) = 0$, $P_1(y) = 1$, and $P_m(y) = P_{m-1}(y) + y P_{m-2}(y)$ for $m \geq 2$, Pascal’s identity $\binom{n-k+2}{k} = \binom{n-k+1}{k} + \binom{n-k+1}{k-1}$ gives $f_n(y) = P_{n+3}(y)$: the two sides satisfy the same three-term recursion in n , and agree at $n = -2, -1$. The Pascal step is machine-checked (`NdimRate.lean`, `pascal_step` and `coeff_pascal`; `pascal_step` depends on *no* axioms), so the only unformalised part of this paragraph is the induction that assembles it. The characteristic equation $x^2 = x + y$ has roots $\alpha, \beta = (1 \pm \sqrt{1+4y})/2$, and the Binet formula yields $P_m(y) = (\alpha^m - \beta^m)/(\alpha - \beta)$; hence $P_m(y) = 0$ iff $(\alpha/\beta)^m = 1$. Setting $\alpha/\beta = e^{2\pi i j/m}$ and solving for y :

$$y_j = -\frac{1}{4 \cos^2(\pi j/m)}, \quad j = 1, 2, \dots, \lfloor (m-1)/2 \rfloor.$$

Translating back via $y = -1/(1+t)$ and using $4 \cos^2 \theta - 1 = 1 + 2 \cos(2\theta)$:

$$t_j = 4 \cos^2\left(\frac{\pi j}{n+3}\right) - 1 = 1 + 2 \cos\left(\frac{2\pi j}{n+3}\right), \quad 1 \leq j \leq \lfloor (n+2)/2 \rfloor.$$

The asymptotic growth rate is the largest root, attained at $j = 1$. □ □

Remark 3 (Prior art). Both ingredients of Theorem 2 are classical and are used as such. Steinberg’s formula for the Poincaré series of a Coxeter group is standard, and the Fibonacci–Pell polynomials $P_m(y)$, their Binet formula, and the location of their roots at $y_j = -1/(4 \cos^2(\pi j/m))$ are classical facts about that family; the values $1 + 2 \cos(2\pi j/m)$ are likewise the familiar Chebyshev-type quantities. What is asserted here is the *identification*: that the abstract orbit-growth rate of the n -dimensional right-corner orthoscheme group is the largest of those roots at $m = n + 3$, together with the collision-depth invariant of §4. We have not located that identification in the literature, but we make no claim of novelty for the ingredients, and a reader who knows the Coxeter growth-series literature should treat the closed form as an assembly of known pieces rather than a new evaluation.

Remark 4 (The Class C orthoscheme sequences in the OEIS). For pairwise distinct legs the layer counts u_d^n are recorded in the On-Line Encyclopedia of Integer Sequences: the planar base case ($n = 2$) is A396406, and the orthoschemes are A397439 ($n = 3$), A397437 ($n = 4$), A396927 ($n = 5$), and A397438 ($n = 6$), with the $n = 7$ entry in preparation. Each is a rational generating function whose dominant pole is $1/r_n$ for r_n of Theorem 2; for instance $(1+x)^2/(1-2x)$ at $n = 3$, $(1+x)^3/(1-2x-x^2+x^3)$ at $n = 4$, $(1+x)^4/(1-3x+3x^3)$ at $n = 6$, and $(1+x)^4/(1-4x+3x^2+2x^3-x^4)$ at $n = 7$. The planar case $n = 2$ and its lamplighter structure are treated in a companion paper.

Corollary 5 (asymptotic regime). $r_n \rightarrow 3$ as $n \rightarrow \infty$, with the explicit residual

$$3 - r_n = 4 \sin^2\left(\frac{\pi}{n+3}\right) \sim \frac{4\pi^2}{(n+3)^2}.$$

The closed form gives quadratic-surd values exactly when $\varphi(n+3) \leq 4$ (Euler’s totient), namely $n+3 \in \{5, 6, 8, 10, 12\}$; see Table 1.

n	r_n	algebraic form
2	1.6180...	$\varphi = (1 + \sqrt{5})/2$ (golden)
3	2.0000	exactly 2
4	2.2470...	$1 + 2 \cos(2\pi/7)$, cubic
5	2.4142...	$1 + \sqrt{2}$ (silver)
6	2.5321...	$1 + 2 \cos(2\pi/9)$, cubic
7	2.6180...	$\varphi^2 = (3 + \sqrt{5})/2$
8	2.6825...	$1 + 2 \cos(2\pi/11)$, degree 5
9	2.7321...	$1 + \sqrt{3}$
10	2.7709...	$1 + 2 \cos(2\pi/13)$, degree 6

Table 1: The closed-form rate $r_n = 1 + 2 \cos(2\pi/(n+3))$ for the right-corner orthoscheme reflection group, verified to 50-digit precision for n up to 500.

3. Unconditional Class C faithfulness

The Steinberg series (2) is an upper bound: only when the affine representation $\rho_n : W_n \rightarrow \text{Aff}(\mathbb{R}^n)$ is faithful and no additional Coxeter relation closes does the BFS orbit count u_d^n saturate $[t^d] W_n(t)$. Following the terminology of the 3D analysis, we call a leg tuple $(a_1, \dots, a_n)$ *Class C* when its coordinates are pairwise distinct. The main result of this section certifies that the Class C case is unconditional in every dimension.

3.1 The uniform Schur-complement determinant identity

The Lorentzian Gram matrix of the orthoscheme in staircase coordinates (with simple roots $n_0 = e_1$, $n_i = a_i e_{i+1} - a_{i+1} e_i$ ($1 \leq i \leq n-1$), $n_n = e_n$, and the standard $(n, 1)$ endpoint

correction) is the tridiagonal matrix Q_n of size $(n+1) \times (n+1)$ given by

$$\begin{aligned} Q_n[0,0] &= 1 - a_1^2, & Q_n[i,i] &= a_i^2 + a_{i+1}^2 \quad (1 \leq i \leq n-1), & Q_n[n,n] &= 1, \\ Q_n[0,1] &= a_2, & Q_n[i,i+1] &= a_i a_{i+2} \quad (1 \leq i \leq n-2), & Q_n[n-1,n] &= a_{n-1}. \end{aligned}$$

Lemma 6 (Uniform Schur-complement determinant identity). *For every $n \geq 3$ and every positive tuple $(a_1, \dots, a_n)$,*

$$\det Q_n = - \prod_{i=1}^n a_i^2.$$

Proof. Let $D_k = \det Q_n[:k+1, :k+1]$ be the leading principal minor of order $k+1$. Tridiagonal expansion along the last row gives

$$D_k = Q_n[k,k] D_{k-1} - Q_n[k-1,k]^2 D_{k-2} \quad (1 \leq k \leq n), \quad (\dagger)$$

with $D_{-1} = 1$. We claim

$$D_k = P_k := \left(\prod_{i=1}^k a_i^2 \right) \left(1 - \sum_{i=1}^{k+1} a_i^2 \right) \quad \text{for every } 1 \leq k \leq n-1. \quad (\star)$$

Base case. $D_0 = 1 - a_1^2$, and $D_1 = (1 - a_1^2)(a_1^2 + a_2^2) - a_2^2 = a_1^2(1 - a_1^2 - a_2^2) = P_1$.

Interior step (uniform). For $2 \leq k \leq n-1$ the recurrence $(\dagger)$ reads $D_k = (a_k^2 + a_{k+1}^2) D_{k-1} - (a_{k-1} a_{k+1})^2 D_{k-2}$. Introduce the abstract symbols $S_{k-1} := \sum_{i=1}^{k-1} a_i^2$ and $P_{k-2} := \prod_{i=1}^{k-2} a_i^2$; substituting the closed forms $(\star)$ reduces the step to the polynomial identity

$$\begin{aligned} P_{k-2} a_{k-1}^2 a_k^2 (1 - S_{k-1} - a_k^2 - a_{k+1}^2) &= (a_k^2 + a_{k+1}^2) P_{k-2} a_{k-1}^2 (1 - S_{k-1} - a_k^2) \\ &\quad - (a_{k-1} a_{k+1})^2 P_{k-2} (1 - S_{k-1}) \end{aligned}$$

in the five independent symbols $(a_{k-1}, a_k, a_{k+1}, S_{k-1}, P_{k-2})$. This identity is independent of k and of n . It, the two base cases, the induction that assembles them into $(\star)$, and the boundary step yielding $\det Q_n$, are machine-checked in Lean 4 over an arbitrary commutative ring: see `OrthoschemeDet.lean`, theorem `det_Q_of_recurrence` (Section 7). No positivity, ordering, or characteristic hypothesis is used.

Final step (uniform). At $k = n$ the recurrence uses $Q_n[n,n] = 1$ and $Q_n[n-1,n]^2 = a_{n-1}^2$, giving $D_n = D_{n-1} - a_{n-1}^2 D_{n-2}$. With $S_{n-1} = \sum_{i=1}^{n-1} a_i^2$, $P_{n-2} = \prod_{i=1}^{n-2} a_i^2$,

$$D_{n-1} - a_{n-1}^2 D_{n-2} = P_{n-2} a_{n-1}^2 [(1 - S_{n-1} - a_n^2) - (1 - S_{n-1})] = -P_{n-2} a_{n-1}^2 a_n^2 = - \prod_{i=1}^n a_i^2.$$

This final identity is again a single polynomial identity in the four independent symbols $(a_{n-1}, a_n, S_{n-1}, P_{n-2})$, independent of n . The three identities (base, interior, final) jointly close the induction for every $n \geq 3$, with no per- n verification required. $\square$ $\square$

3.2 Diophantine emptiness on three Cremona curves

A non-perpendicular dihedral pair $(i, i+1)$ in the staircase orthoscheme has $\cos^2 \theta_{i,i+1} \in \mathbb{Q}$; by the extended Niven theorem (a consequence of Conway–Jones [4]: if $\theta \in \mathbb{Q}\pi$ and $\cos^2 \theta \in \mathbb{Q}$, then $\cos \theta \in \{0, \pm 1/2, \pm \sqrt{2}/2, \pm \sqrt{3}/2, \pm 1\}$), any non-trivial closing relation $(R_i R_{i+1})^m = 1$ forces $\cos^2 \theta_{i,i+1} \in \{0, 1/4, 1/2, 3/4, 1\}$. The boundary values $\{0, 1\}$ are immediately incompatible with positive distinct legs. The remaining three cut out, in the two legs flanking the interior pair (with $X = a_i^2$, $Z = a_{i+2}^2$), the Diophantine quartics

$$\mathcal{V}_{1/4} : X^2 + 14XZ + Z^2 = m^2, \quad \mathcal{V}_{1/2} : X^2 + 6XZ + Z^2 = k^2, \quad \mathcal{V}_{3/4} : 9X^2 + 30XZ + 9Z^2 = k^2.$$

Lemma 7. *Each of $\mathcal{V}_{1/4}$, $\mathcal{V}_{1/2}$, $\mathcal{V}_{3/4}$ is empty over distinct-coordinate $\mathbb{Q}_{>0}^2$ (i.e., the only rational solutions have $XZ = 0$ or $X = Z$).*

Proof. $\mathcal{V}_{1/2}$. The quartic $X^2 + 6XZ + Z^2 = k^2$ is birationally equivalent to the elliptic curve $E_6 : y^2 = x^3 - x$ (Mordell [5], §17). E_6 has Mordell–Weil rank 0 over $\mathbb{Q}$ by Fermat’s classical descent (Cassels [6], Ch. II), and its full rational-point set is the 2-torsion $\{\infty, (0, 0), (\pm 1, 0)\}$, pulling back to $XZ = 0$ or $X = Z$.

$\mathcal{V}_{1/4}$. The Jacobian elliptic curve is

$$E_{1/4} : y^2 = x(x - 12)(x - 16),$$

which is Cremona 24a1 (conductor 24, $j = 35\,152/9$, Mordell–Weil rank 0 over $\mathbb{Q}$, torsion $\mathbb{Z}/4 \oplus \mathbb{Z}/2$); see [7] and the LMFDB record [8]. Each of its eight rational points pulls back to $XZ = 0$ or $X = Z$.

$\mathcal{V}_{3/4}$. The quartic $9X^2 + 30XZ + 9Z^2 = k^2$ admits a standard conic-to-Weierstrass substitution and reduces to

$$E_{3/4} : y^2 = x(x - 300)(x - 1200),$$

Cremona 72a2 (conductor 72, same j -invariant $35\,152/9$ as $E_{1/4}$ (they are quadratic twists) and also rank 0 with torsion $\mathbb{Z}/2 \oplus \mathbb{Z}/2$); see [7, 9]. Each torsion point again pulls back to a trivial quartic solution. $\square$ $\square$

3.3 Faithfulness theorem

Theorem 8 (Class C faithfulness, all $n \geq 3$, unconditional). *For every $n \geq 3$ and every leg tuple $(a_1, \dots, a_n) \in \mathbb{Q}_{>0}^n$ with pairwise distinct coordinates, the affine representation $\rho_n : W_n \rightarrow \text{Aff}(\mathbb{R}^n)$ is faithful, no Coxeter relation closes beyond the $\binom{n}{2} - n$ perpendicular pairs, and the BFS layer counts saturate the abstract Coxeter sphere size:*

$$u_d^n(O) = [t^d] W_n(t) \quad \text{for every } d \geq 0.$$

Proof. By Lemma 6, $\det Q_n = -\prod_{i=1}^n a_i^2 \neq 0$ for every positive tuple, so the Lorentzian Coxeter system is non-degenerate. Tits’ theorem on faithfulness of the geometric representation (Humphreys [1], Thm. 5.4) then gives faithfulness of ρ_n on every positive-rational leg tuple.

It remains to exclude additional Coxeter relations. Each non-perpendicular pair is of type $(i, i + 1)$, $0 \leq i \leq n - 1$; its cosine satisfies $\cos^2 \theta_{i,i+1} \in \mathbb{Q}$ (boundary pairs $(0, 1)$ and $(n - 1, n)$ involve two legs, interior pairs involve three). By the extended Niven theorem, a closing Coxeter relation requires $\cos^2 \theta_{i,i+1} \in \{0, 1/4, 1/2, 3/4, 1\}$. The boundary values $\{0, 1\}$ contradict positivity of legs. At boundary pairs, the values $1/4, 3/4$ force $a_i = a_{i+1}/\sqrt{3}$ or $a_i = a_{i+1}$, the first irrational and the second excluded by the distinct-coordinate hypothesis; the value $1/2$ forces $a_i = a_{i+1}$, again excluded. At each interior pair, the three values $1/4, 1/2, 3/4$ define exactly the three Diophantine quartics $\mathcal{V}_{1/4}, \mathcal{V}_{1/2}, \mathcal{V}_{3/4}$ in the two flanking legs (a_i, a_{i+2}) (the dihedral analysis is uniform in n). By Lemma 7 each of the three is empty over distinct-coordinate $\mathbb{Q}_{>0}^2$.

Hence outside the symmetry locus $\{a_i = a_{i+1}\}$ for some i , no non-perpendicular dihedral is a rational multiple of π , no closing relation appears, and $u_d^n(O) = [t^d] W_n(t)$. $\square$ $\square$

4. Dimension-independent collision-depth

For a leg sequence $(a_1, \dots, a_n)$ define the *collision-depth* as

$$\text{cd}_n(a_1, \dots, a_n) = \inf \{ d \geq 0 : u_d^n < [t^d] W_n(t) \} \in \{1, 2, 3, \dots, \infty\}.$$

By Theorem 8, $\text{cd}_n = \infty$ for every pairwise-distinct tuple. The next theorem extends the description to every leg multiplicity pattern and shows that, in spite of the multiplicity partition having $p(n)$ possible shapes, the collision-depth takes only three values, determined by two local positional statistics.

For $(a_1, \dots, a_n) \in \mathbb{Q}_{>0}^n$ set

$$\begin{aligned} e(a_1, \dots, a_n) &= \#\{i \in \{1, n-1\} : a_i = a_{i+1}\} \in \{0, 1, 2\}, \\ t(a_1, \dots, a_n) &= \#\{i : 1 \leq i \leq n-2, a_i = a_{i+1} = a_{i+2}\}. \end{aligned}$$

Thus e counts equal pairs at the two *endpoint* (boundary) positions of the staircase, and t counts *interior triples* of three consecutive equal legs. An equal pair sitting strictly in the interior of the staircase (such as $a_2 = a_3$ in $(1, 2, 2, 3)$ at $n = 4$) contributes to neither e nor t .

Theorem 9 (Dimension-independent collision-depth, all $n \geq 3$). *For every $n \geq 3$ and every leg sequence $(a_1, \dots, a_n) \in \mathbb{Q}_{>0}^n$,*

$$\text{cd}_n(a_1, \dots, a_n) = \begin{cases} \infty & \text{if } e = t = 0, \\ 4 & \text{if } e \geq 1, t = 0, \\ 3 & \text{if } t \geq 1. \end{cases}$$

In particular, $\text{cd}_n \in \{\infty, 4, 3\}$ for every n , and the value depends only on the local positional statistics (e, t) .

Proof. The non-perpendicular dihedral pairs of $O = O(a_1, \dots, a_n)$ are (R_i, R_{i+1}) , $0 \leq i \leq n-1$. Their cosines have the following uniform form:

$$\begin{aligned} \cos \theta_{0,1} &= \frac{a_2}{\sqrt{a_1^2 + a_2^2}}, & \cos \theta_{n-1,n} &= \frac{a_{n-1}}{\sqrt{a_{n-1}^2 + a_n^2}}, \\ \cos \theta_{i,i+1} &= \frac{a_i a_{i+2}}{\sqrt{(a_i^2 + a_{i+1}^2)(a_{i+1}^2 + a_{i+2}^2)}} & & \text{for } 1 \leq i \leq n-2. \end{aligned}$$

Endpoint pair $\Rightarrow \text{cd} = 4$. An endpoint pair $a_1 = a_2$ (or $a_{n-1} = a_n$) makes the boundary cosine $1/\sqrt{2}$, so the dihedral is $\pi/4$ and the Coxeter relation $(R_0 R_1)^4 = 1$ (resp. $(R_{n-1} R_n)^4 = 1$) closes. The parabolic $\langle R_0, R_1 \rangle$ then has order 8 rather than infinite, and the first collision in the orbit BFS appears at length 4, giving $\text{cd}_n = 4$.

Interior triple $\Rightarrow \text{cd} = 3$. An interior triple $a_i = a_{i+1} = a_{i+2}$ ($1 \leq i \leq n-2$) makes the interior cosine $\cos \theta_{i,i+1} = a^2/(2a^2) = 1/2$ where $a := a_i$, so the dihedral is $\pi/3$ and the braid relation $R_i R_{i+1} R_i = R_{i+1} R_i R_{i+1}$ closes at length 3. Since a triple subsumes two pairs, the $\pi/3$ relation appears before any $\pi/4$ relation that the same triple might also induce at a boundary position, giving $\text{cd}_n = 3$.

Interior pair only $\Rightarrow \text{cd} = \infty$. Suppose $(e, t) = (0, 0)$, but the sequence is not Class C; then some interior pair $a_j = a_{j+1}$ exists with $1 \leq j \leq n-1$, neither boundary position nor part of a triple. At each non-perpendicular pair $(i, i+1)$ the cosine depends on the flanking legs, which are by hypothesis not all equal. At the two interior pairs flanking the equal pair $((j-1, j)$ and $(j+1, j+2)$ (whichever exist)) the cosine takes the $a_i a_{i+2} / \sqrt{(a_i^2 + a_{i+1}^2)(a_{i+1}^2 + a_{i+2}^2)}$ form with (a_i, a_{i+2}) distinct positive rationals (otherwise the equal pair would extend to a triple, contradicting $t = 0$). By the extended Niven theorem and Lemma 7, no closing relation is possible: each of the candidate values $1/4, 1/2, 3/4$ for $\cos^2$ defines one of the three Diophantine quartics, all empty over distinct $\mathbb{Q}_{>0}^2$. At every non-flanking pair the flanking-leg analysis is the same. Hence no Coxeter relation closes, and by Tits faithfulness on the resulting non-degenerate Coxeter system (Lemma 6 plus Humphreys [1], Thm. 5.4) $u_d^n = [t^d] W_n(t)$ for every d , i.e., $\text{cd}_n = \infty$.

Combining the three cases proves the tri-classification. $\square$ $\square$

Remark 10 (the R -rule fails in $n \geq 4$). A natural but incorrect heuristic is to set cd_n to a finite value as soon as some consecutive equal pair appears, i.e., as soon as the run-length $R := \max_i \#\{j : a_j = a_i\}$ is at least 2. At $n = 3$ this R -rule coincides with the (e, t) -rule because every consecutive equal pair in a 3-term sequence is automatically at an endpoint. At $n \geq 4$ the two rules diverge: the orthoscheme $(1, 2, 2, 3)$ has an interior equal pair ($R = 2$, but $(e, t) = (0, 0)$) and Theorem 9 predicts $\text{cd}_4 = \infty$; empirically the BFS sequence $1, 5, 14, 33, 75, 169, \dots$ is bit-identical to the Class C sequence $1, 5, 14, 33, 75, 169$ at $(1, 2, 3, 5)$ through every observed depth, falsifying the R -rule.

5. Atlas: small dimensions and the (e, t) classes

We collect a small atlas of representative leg sequences for $n = 3, 4, 5, 6$, illustrating each attainable (e, t) class. The layer counts are exact-rational breadth-first search; in every case the observed collision depth agrees with the value cd_n predicted by Theorem 9.

n	leg sequence	(e, t)	cd_n	first six BFS layer counts
3	(2, 3, 5)	(0, 0)	∞	1, 4, 9, 18, 36, 72
3	(1, 1, 2)	(1, 0)	4	1, 4, 9, 18, 35, 67
3	(1, 1, 1)	(2, 1)	3	1, 4, 9, 17, 28, 42
4	(1, 2, 3, 5)	(0, 0)	∞	1, 5, 14, 33, 75, 169
4	(1, 2, 2, 3)	(0, 0)	∞	1, 5, 14, 33, 75, 169
4	(1, 1, 2, 3)	(1, 0)	4	1, 5, 14, 33, 74, 163
4	(1, 1, 2, 2)	(2, 0)	4	1, 5, 14, 33, 73, 157
4	(1, 1, 1, 2)	(1, 1)	3	1, 5, 14, 32, 67, 135
5	(1, 2, 3, 5, 7)	(0, 0)	∞	1, 6, 20, 54, 136, $\dots$
5	(1, 2, 2, 3, 5)	(0, 0)	∞	1, 6, 20, 54, 136, $\dots$
5	(1, 1, 2, 3, 5)	(1, 0)	4	1, 6, 20, 54, 135, 327
5	(2, 1, 1, 1, 2)	(0, 1)	3	1, 6, 20, 53, 128, 297
6	(1, 1, 2, 3, 2, 2)	(2, 0)	4	1, 7, 27, 82, 224, 581

Table 2: A small atlas of representative orthoschemes in $n = 3, 4, 5, 6$ illustrating each (e, t) class. The collision-depth value $\text{cd}_n \in \{\infty, 4, 3\}$ is determined uniformly by (e, t) in every dimension, by Theorem 9. The two $n = 4$ sequences with $(e, t) = (0, 0)$ have bit-identical BFS counts through every observed depth despite differing in run structure ($(1, 2, 3, 5)$ has $R = 1$ but $(1, 2, 2, 3)$ has $R = 2$; only (e, t) is the right invariant).

The all-distinct rows of the table saturate the Steinberg upper bound $[t^d] W_n(t)$ by Theorem 8; for $n = 3$ this reproduces the classical sequence $u_d = 9 \cdot 2^{d-2}$ for $d \geq 2$.

6. Related work and open questions

Path Coxeter groups and the Cannon–Floyd–Parry framework. The closed form $r_n = 1 + 2 \cos(2\pi/(n+3))$ is the structural signature of a linear (path-shaped) Coxeter group whose Poincaré-series denominator reduces to a tridiagonal Chebyshev-type characteristic polynomial. Cannon and Wagreich [10] and Floyd and Plotnick [11] developed the growth-rate calculus of such groups; Theorem 2 fits into their framework with parameter $m = n + 3$ and is recovered cleanly through the Fibonacci–Pell substitution of §2.

Salem-type family on the surface-dynamics side. The family of algebraic integers $\{1 + 2\cos(2\pi/m)\}_{m \geq 3}$ to which r_n belongs also arises in birational surface dynamics. Diller and Favre [13] introduced the dynamical-degree framework for bimeromorphic maps of surfaces; McMullen [14] showed that this family arises as dynamical degrees of automorphisms of blowups of $\mathbb{P}^2$ associated with the E_n T-shaped Coxeter graph (different from the path Coxeter graph here); and Bedford–Kim [15] realised specific members through linear-fractional recurrences. The rate r_n lies in this family, but the right-corner orthoscheme reflection group is not known to be realised as a McMullen involution-product on a rational surface; the coincidence is at the level of the algebraic-number family. A birational-geometry construction realising r_n as a dynamical degree on a $\mathbb{P}^2$ -blowup would be of independent interest.

Open questions. (1) *Non-orthoscheme reflection groups.* Does an analogue of Theorem 9 hold for other families of simplicial reflection chambers in $\mathbb{R}^n$: for example, Coxeter orthoschemes with bent paths, or simplices arising from non-path-shaped Coxeter diagrams? The proof relies on the right-angled-with-one- ∞ structure of the path Coxeter diagram; how robust is the (e, t) -classification under deformation of the diagram?

(2) *Classes of intermediate growth.* In every dimension $n \geq 3$, partial multiplicity patterns (neither distinct nor saturated by triples) produce intermediate growth rates strictly between two consecutive values of r_n . An algebraic characterisation of these intermediate rates (in particular, whether they are algebraic over $\mathbb{Q}$ at all) is open.

(3) *Beyond (e, t) .* Theorem 9 shows that the collision-depth is determined by the maximum multiplicity of *positional* structure (e, t) . Is there a finer positional invariant that captures the entire u_d^n sequence (not only cd_n) in terms of $(a_1, \dots, a_n)$?

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7. Machine verification

The algebraic content of Lemma 6 and of Theorem 2 is formalised in Lean 4 with Mathlib. Each file ends with `#print axioms`; every theorem below reports only the standard axioms `propext`, `Classical.choice`, `Quot.sound` (two report strictly fewer), and no file contains `sorry`.

Paper result	Lean declaration	File
Lemma 6, closed form ($\star$)	<code>minor_closed_form</code>	<code>OrthoschemeDet.lean</code>
Lemma 6, boundary step	<code>det_Q</code>	<code>OrthoschemeDet.lean</code>
Lemma 6, full statement	<code>det_Q_of_recurrence</code>	<code>OrthoschemeDet.lean</code>
Thm. 2, root of $x^2 = x + y$	<code>sq_eq</code>	<code>NdimRate.lean</code>
Thm. 2, Binet formula	<code>pell_binet</code>	<code>NdimRate.lean</code>
Thm. 2, root translation	<code>root_translation</code>	<code>NdimRate.lean</code>
Corollary (asymptotic regime)	<code>three_sub_r</code>	<code>NdimRate.lean</code>
Thm. 2, Pascal step	<code>pascal_step</code> (no axioms)	<code>NdimRate.lean</code>
Thm. 2, coefficient form	<code>coeff_pascal</code>	<code>NdimRate.lean</code>

What is not formalised. The step identifying the abstract orbit-growth rate with the largest root t_1 of W_n is a dominant-pole argument on the rational generating function; it is proved on paper only. Likewise the combinatorial identification $f_n(y) = P_{n+3}(y)$ via Pascal’s identity is a paper proof. Everything downstream of those two inputs (the Binet formula, the location of the roots, and the translation to $1 + 2\cos(2\pi j/(n+3))$) is machine-checked.

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